MEZO BUILDATHON · TRACK 1 — DEFI

Pylon

Non-custodial flash leverage, self-deleveraging and a redemption shield for Bitcoin on Mezo — on your own trove, with one signature.

TESTNET 31611

Full product, live: engine, reserve, position, terminal.

MAINNET 31612

Same audited code, funded with real MUSD, migrated to a new generation.

PROOF

43 contract tests · 12-lens audit · every claim resolves to a transaction.

pylon-mezo.vercel.app

github.com/PhiBao/pylon-mezo

THE TRAP

A Mezo loan is static, and the queue is invisible.

1. **Looping is manual.** Borrow → swap → re-collateralise, 6–15 transactions, each moving your ratio.
2. **Repayment needs MUSD you may not have.** Collateral is released only *after* the repayment. A borrower with equity and a falling price cannot reach it.
3. **Liquidation is not the only risk.** Redemptions walk up from the lowest ICR above 110%. A 135% borrower loses BTC with no liquidation warning.

429M MUSD of redemption depth behind 296 testnet troves — and no tooling to steer around it.

THE INSIGHT

Mezo already contains every part. Nobody composed them.

BorrowerOperationsSignatures

A contract can act on your existing trove from one EIP-712 signature. No custody, no migration.

Tigris pools

Solidly-style `hook` flash liquidity, inside the DEX you already trade on.

The BTC ledger

One 18-decimal native/ERC-20 ledger, so BTC bought in a swap pays the next call's value directly.

That composition is the product: three atomic paths and a priced defence market, built on primitives Mezo already ships.

WHAT IT DOES

Four operations. One transaction each.

MODE	WHAT HAPPENS
Flash Loop	Borrow, swap to BTC, post it as collateral, repay the flash out of the trove's new borrow.
Flash Unwind	Repay debt out of the trove's **own collateral** — with zero MUSD in the wallet.
Flash Exit	Close the trove entirely, funded by itself, and return the rest as BTC.
Redemption Shield	A MUSD reserve that only ever repays a shielded position's debt, within hard caps, with the claim enforced in code.

The signature carries the exact collateral and debt deltas. If a swap cannot deliver them, the whole transaction reverts.

PROOF, NOT PROMISES

Live on both chains.

AUDITED GENERATION · TESTNET

Flash loop	0xc1517166
Flash unwind	0x4a2d139a
Shield defence	0xa4fb8061
Keeper defence	0x456e815b

0.0457 BTC at 181.7% ICR · 332.18 MUSD lent · 2.62 MUSD fees · share price 1.002370.

MAINNET · CHAIN 31612

Deployed	engine 0x8DEa4756
Funded	10.19 MUSD
Deposit priced	exactly as paid
Pool migrated	capital intact
Machine payment	0.001 MUSD · x402

A second depositor's shares are worth exactly what they paid — the live proof of the audit's highest-severity fix.

WHY THIS IS HARD

Constraints only live infrastructure reveals.

1. **A Solidly pool cannot be swapped inside its own flash callback** — two venues, by design.
2. **The signature helper burns from its own caller** — keep the flash proceeds in the engine.
3. **The BTC ledger is one balance** — value is already credited when the callee body runs.
4. **Hint resolution is expensive** — the solver supplies hints, the contract validates them.

We found these by executing them — then had twelve adversarial lenses look for what we missed: 35 findings, 32 fixed, each with a regression test.

THE SHIELD ECONOMY

A priced, enforceable market for redemption defence.

1. Borrowers arm a band: **defend below X% ICR, toward Y%.**
2. Anyone can defend — the owner, a keeper, a competing bot.
3. The reserve lends MUSD that can **only** repay that position's debt.
4. The claim blocks collateral release until it is settled.

0.20%

origination fee on every flashed notional — lender yield

181.7%

the live shielded position, after an autonomous keeper defence

TRACK RECORD IN THIS BUILD

Everything a judge can check.

43

Foundry tests over a faithful Mezo/Tigris harness, including a BTC mock that models the unified native ledger.

9

Rust tests with vectors cross-checked byte-for-byte against `cast` (ABI + EIP-712).

2

chains deployed: full product on testnet, funded generation on mainnet with a live migration.

A Rust engine (radar, solver, keeper, x402, relay) on Fly. A terminal on Vercel that works with no backend at all. Escape routes proven with real value: depositors withdraw, positions settle, and the pool migrated to a new generation with nothing stranded.

BEYOND THE HACKATHON

The MUSD economy is the market.

Mezo mainnet lists 33 MUSD pairs — BTC, mUSDC, mUSDT, WUSD, sUSDat, mSolvBTC, and the tokenised-equity pairs (MSTR, COIN, TSLA, NVDA, GOOGL, CRCL, BTDR) that make Mezo different. Every one has the same redemption risk and no equivalent tool.

The engine takes its venues as constructor arguments, so serving a new pair is a deployment, not a rewrite. Next: defence ballots for native troves, multi-leg yield routing, a keeper bounty market, and an external audit before third-party capital.

CLOSING

A Bitcoin loan should be a position you can steer, not a contract you obey.

TERMINAL
pylon-mezo.vercel.app

ENGINE
pylon-mezo-api.fly.dev

REPOSITORY
github.com/PhiBao/pylon-mezo

MIT